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Started	Thursday, 29 October 2026, 7:07 PM
Completed	Thursday, 29 October 2026, 7:12 PM
Duration	5 mins 3 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Quick Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,80,90,150,200,300,400,470,70,120
b. 200,470,80,150,40,90,300,400,70,120
c. 80,150,200,470,40,90,300,400,70,120
d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data sequentially
c. Sorting data in a single pass
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently
b. Decreases the efficiency of algorithms
c. Makes code execution slower
d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Removing duplicates from a list
b. Merging two datasets
c. Arranging data in a specific order
d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Backtracking
- c. Greedy method
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Complexity
- c. Merging
- d. Searching

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Bubble Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. sorted()
- c. arrange()
- d. order()

The correct answer is: sort()

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It builds a sorted array one element at a time
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It requires external libraries
- b. It only works with integer arrays
- c. It sorts data out of the box efficiently
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Arranging data in a particular format
- c. Searching for data in a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Being the most efficient sorting algorithm
- b. Bubbling up the largest element to its correct position with each pass
- c. Sorting data in a non-sequential manner
- d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching

b. Merging

c. Sorting

d. Rearranging

The correct answer is: Sorting